SAFA

MODULE 4 | AIRCRAFT FLIGHT CONTROLS

ASYS 245 • Part B • Cornell Notes • SECTION 1

Stabilator • anti-servo tabs • trim • aerodynamic balancing

CUES / QUESTIONS

1. What moves when a stabilator controls pitch?

2. What does an anti-servo tab do?

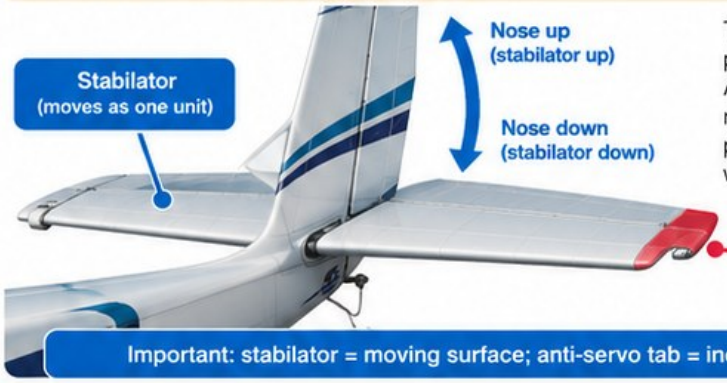
3. How does an elevator trim tab work?

4. What are manual and electric trim controls?

5. How do balance tabs and panels help?

NOTES | STABILATORS AND CONTROL TABS

1 STABILATOR + ANTI-SERVO TAB



The entire horizontal stabilizer pivots as one unit to control pitch. An anti-servo tab increases resistance and control feel, helping prevent overcontrol. Balance weights may also be used.

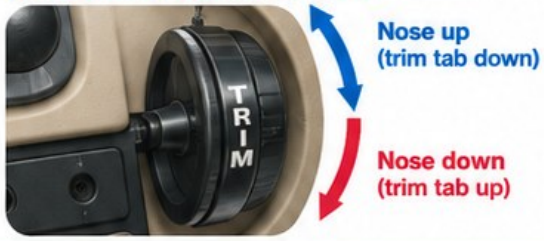
Important: stabilator = moving surface; anti-servo tab = increased control feel.

2 TRIM TABS + CONTROL METHODS



An upward trim tab deflects local airflow upward, tending to move the elevator downward and produce nose-down trim. The tab reduces sustained pilot control force in a selected flight condition.

SMALL AIRCRAFT: manual trim wheel

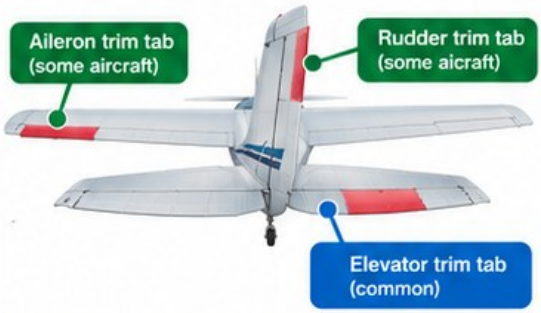


LARGER AIRCRAFT: electric motor / switch



3 WHERE IS TRIM USED?

Commonly on the elevator, but some aircraft also have aileron or rudder trim tabs.



4 BALANCE TABS + PANELS

Aerodynamic forces assist movement of a control surface when control loads are high. Balance panels use differential pressure; balance tabs are linked to the control surface.



SECTION 1 | STUDENT SUMMARY

- Stabilators pivot as a whole to change pitch; anti-servo tabs add control feel.
- Trim can act on elevator, aileron or rudder, depending on the aircraft.
- Trim reduces the force needed to maintain a selected flight condition.
- Balance tabs and pressure panels reduce heavy aerodynamic control loads.